

Day 1: HTML CSS

Topic 1: How web works

Real World	Web Equivalent
You read menu	Browser loads HTML
You place order	HTTP request
Kitchen cooks	Server processes
Waiter delivers	HTTP response
You see food	Browser renders UI

 What Happens When You Type a URL?

1 URL Parsing (Understanding the Address)

Example: `https://www.example.com/page`

The browser breaks it into:

- **Protocol** → `https` (how to communicate)
- **Domain** → `www.example.com` (where to go)
- **Path** → `/page` (what to request)

2 DNS Lookup (Finding the Server's IP)

Computers don't understand domain names — they use **IP addresses**.

So, the browser asks the **DNS (Domain Name System)**:

“What is the IP of `www.example.com`?”

DNS responds with something like:

142.250.183.206

TCP Connection — The 3-Way Handshake (How Two Machines Agree to Talk)

Before any website data is sent, your browser must create a **reliable connection** with the server.

TCP (Transmission Control Protocol) is designed to guarantee:

- ✓ Data arrives
- ✓ Data arrives in order
- ✓ No data is duplicated
- ✓ Lost data is retransmitted

To achieve this, TCP performs a **3-step negotiation** called the **Three-Way Handshake**.

How They Work Together in the Handshake

SYN → Synchronise , **ACK** → Acknowledge

Step	Flag Used	Meaning
1	SYN	Client wants to start connection
2	SYN + ACK	Server accepts and responds
3	ACK	Client confirms → Connection established

Why Are They Important?

These flags ensure:

- Both sides are ready before sending data
- Data transfer can be tracked reliably
- Lost packets can be detected and resent
- Communication is synchronized

Without SYN and ACK, computers would send data blindly — like talking without knowing if anyone is listening.

HTTP Request Sent and Server Responds

GET / HTTP/1.1

Host: google.com

200 OK

Content-Type: text/html

Browser Parses

- HTML → DOM
- CSS → CSSOM
- JS → Executes

Common Interview Questions

What is DNS?

DNS (Domain Name System) is like the internet's phonebook. It converts a human-readable domain (e.g., google.com) into an **IP address** (e.g., 142.250.xxx.xxx) so browsers can find the correct server.

? Difference Between HTTP and HTTPS?

HTTP	HTTPS
No encryption	Encrypted using TLS
Data can be intercepted	Data is secure & private 🔒
http://	https://
Port 80	Port 443

👉 **HTTPS = HTTP + TLS Encryption**, which protects data from hackers.

? What is Statelessness?

HTTP is stateless, meaning:

Each request is treated as completely new.

The server **does not remember previous requests** unless we use:

- Cookies 🍪
- Sessions
- Tokens (JWT)

Example: Logging in requires a token because HTTP itself doesn't remember you.

Practice Problems

Q. Why Is the First Load Slower Than a Refresh?

The first load is slower because the browser must establish connections and download all resources, while refresh uses cached files and existing connections, avoiding most of that work.

Q. Where does caching happen?

Answer:

- Browser cache
- CDN cache
- Server cache

Q. Your Site Loads in **6s in India** but **1s in the US**. Why?

Answer:

The site is slower in India because requests must travel to a distant origin server, causing high latency. A CDN reduces this by serving content from geographically closer edge servers, minimizing round-trip time.

Q. Explain how React app still depends on HTML.

Answer:

A React app still depends on HTML because the browser initially loads an HTML file "index.html" that contains a root element. React mounts onto that element and dynamically generates and updates the DOM inside it, but it cannot run without that initial HTML structure.



How the Web Works — Manual Inspection

Used Chrome DevTools Network Tab to inspect a real HTTP request.

Test URL: <https://www.delta.exchange/>



Request Sent by Browser

Method: GET

Request Headers

Host: www.delta.exchange

User-Agent: Mozilla/5.0 (Linux; Android 8.0.0; SM-G955U Build/R16NW)

AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Mobile

Safari/537.36

Accept:

text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,
image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7

Accept-Encoding: gzip, deflate, br, zstd

Connection: keep-alive

What This Means

- Browser is requesting the root document.
- It tells the server what formats it supports.



Response From Server

Status Code: 200 OK

Meaning

The server successfully processed the request and returned the webpage.

Timing Breakdown

Phase	Explanation
DNS Lookup	Domain translated to IP address
TCP Connection	3-way handshake established
TLS Handshake	Secure HTTPS encryption setup
TTFB	Server processed request
Content Download	Browser received HTML

Key Learning

First page load is slow because:

- DNS + TCP + TLS must be established.
- No caching exists yet.

Refresh is faster because:

- Connection reused.
- Assets loaded from browser cache.
- Only validation requests sent (304 responses).

Conclusion

A webpage load is not just "fetching HTML".

It is a multi-step distributed system involving:

DNS → TCP → TLS → HTTP → Server Processing → Browser Rendering.

Understanding this helps diagnose:

- Slow websites
- CDN needs

- Backend latency
- Caching strategies